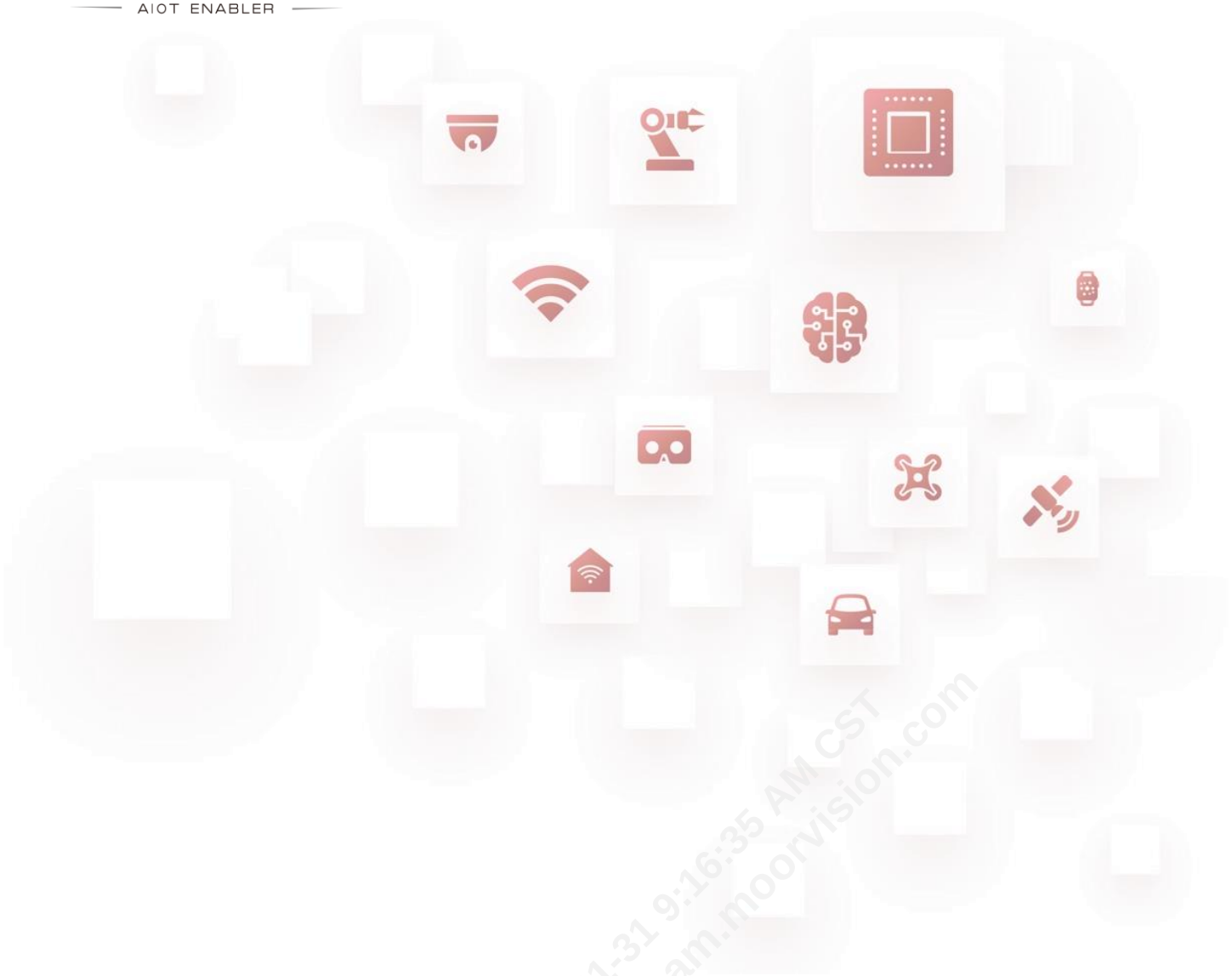
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Thundercomm TurboX SOM Smart Module

Image Flashing User Guide

Rev. V1.2
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Revision History

Revision	Date	Author	Description
1.0	Apr 21, 2023	Lining Jiao	Initial release.
1.1	May 18, 2023	Lining Jiao	Add Chapter 3. Introduction to CDT.
1.2	Sep 19, 2023	Lining Jiao	Change “TurboX Easy Flash” to “Tflash”.

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About This Document

- This document is applicable to all smart modules.
- This document introduces status of the software installed on smart modules before shipment and the necessary flashing procedure for users to perform upon purchase.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.

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Table of Contents

Chapter 1. Factory Status of Smart Module Software ..... - 2 -

Chapter 2. Flashing Methods in Different Scenes ..... - 3 -

2.1. Flashing during R&D stage by customers ..... - 3 -

2.2. Flashing during production by customer ..... - 3 -

2.3. Flashing under the scene of SOM DK ..... - 4 -

Chapter 3. Introduction to CDT ..... - 5 -

Appendix 1. Notices ..... - 6 -

Appendix 2. Trademarks ..... - 8 -

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Chapter 1. Factory Status of Smart Module Software

Software of Thundercomm TurboX smart modules falls into the category of non-standard goods. Generally, after purchasing a smart module, customers need to customize its software. The software installed on the smart module before shipment is the test version of the production line; therefore, it can only be used for module production test and cannot be used on customer products. Customers need to flash the customized software into the smart module upon purchase. Factory data including SN, MAC address, and RF calibration parameters (QCN), etc. had been written into the smart module before shipment. Users shall avoid erasing the factory data during the flashing process.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark: TURBOX.

This document will hereinafter describe flashing methods in different scenes.

- SN and MAC addresses are stored in the **Persist** partition. The **Persist** partition can be erased by the user. Once the **Persist** partition is erased, the SN and MAC addresses will get lost. Thundercomm has modified the storage method of SN and MAC addresses in some modules, and stores them in partitions that are not easily erased. For details, please refer to the relevant documents of each module.

  ➲ **NOTE:** By default, SN and MAC addresses are saved in the **Persist** partition, unless otherwise specified.

- For modules that support WWAN, QCN is stored in fsg, fsc, modemst1, and modemst2 partitions; for modules that do not support WWAN, no QCN is available.

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Chapter 2. Flashing Methods in Different Scenes

2.1. Flashing during R&D stage by customers

In the R&D (Research and Development) stage, customers generally mount the smart module to the IO board, flash the customized software to the module, and complete device bring up and other function verification. Batch operation is unnecessary for flashing in the R&D stage. Users can use Qualcomm's QFIL and PCAT tools, or the Tflash tool provided by Thundercomm to complete the flashing process. Before flashing, it is necessary to back up the factory data first; upon completion of flashing, restore the data. For procedure of data backup and recovery, please refer to the instructions of QFIL, PCAT, and Tflash.

- Due to differences of customer's IO boards, backup of the factory data may not be completed upon bootup. If backup fails, please contact Thundercomm technical support.
- QFIL and PCAT need Qualcomm's authorization, customers without Qualcomm license can use Tflash.

Customers can preferentially flash FastbootBuild; during this process, the factory data will not be erased. As partial factory software is retained during FastbootBuild flashing, great differences between the customer's software/hardware and the SOM production test version, if any, may cause bootup failure or other problems of the system. In this case, it is required to flash FlatBuild. For the specific method of flashing FlatBuild, please refer to the release notes of each SDK.

☞ **FastbootBuild:** An Image format released by Thundercomm; it can be flashed using the Fastboot tool with the precondition that the device has already integrated software and can enter the Fastboot mode.

FlatBuild: An image format released by Thundercomm; it can be flashed in EDL (Emergency download) mode. It is unnecessary to flash the software into the device in advance, Qualcomm's QFIL, PCAT or Thundercomm's Tflash tool are required for flashing.

☞ The customer's IO board **must support flashing in EDL mode**, otherwise the SOM software may not be updated.

2.2. Flashing during production by customer

Batch flashing is generally required for the production line, and the factory data in the module needs to be retained.

Refer to the following methods to retain the factory data.

Method 1. Do not erase the factory data during the flashing process.

Method 2. Back up the factory data before flashing and restore the data upon flashing.

➲ **NOTE:** If the customer's software version is inconsistent with the partition table of the SOM software version at factory status, you can **only use method 2**.

Customers can develop production line flashing tools by themselves, or contact the technical support personnel of Thundercomm to obtain Thundercomm production line flashing tools.

☞ The customer's production line tools **must support flashing in EDL mode**, otherwise the SOM software may not be updated.

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2.3. Flashing under the scene of SOM DK

DK (Development Kit) is a combination of SOM + IO board released by Thundercomm. Customers can purchase DK to evaluate SOM, or perform software development and verification based on DK.

SDK released by Thundercomm is developed based on DK. When the SDK is released, relevant code and image will be released synchronously. The SDK software can be directly flashed to run on the DK. Users can directly flash the image released by Thundercomm to run on the DK, or compile the image based on the SDK code and flash it to run on the DK.

The software installed on the DK before shipment is the production line test version and differs from the software version of the SDK. Upon purchase, customers need to update the software version of the DK to the corresponding SDK version as required. For the update method of the SDK software version, please refer to the release notes of each SDK.

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Chapter 3. Introduction to CDT

CDT is a data block that contains platform/device-specific data, primarily used to enable the same software image to run on different hardware devices.

In general, CDT is written to special partitions on eMMC/UFS during the factory burning process, which are not easily erased. Different hardware devices can be programmed with different CDTs. During system boot-up, the CDT is read in the bootloader stage, and the system software executes different actions based on the configuration information in the CDT, enabling the functionality of running the same software image on different hardware devices.

For Qualcomm platforms, a critical function is to load different DTS (Device Tree Source) files in the kernel stage using the CDT, thus achieving device driver adaptation. It is possible to have devices without a CDT written on Qualcomm platforms. In such cases, the default configuration from the bootloader is used.

The CDT configuration for Thundercomm Compute Modules in their factory state is uncertain. SOM SOMs may not have a CDT written, while others may have a CDT written for testing purposes during the factory stage. Customers cannot directly use the CDT from the compute module; instead, they need to flash the corresponding CDT before flashing their own image. If the customer's software does not require a CDT, the CDT on the SOM needs to be erased. For instructions on flashing and erasing the CDT, please refer to the user guides of QFIL, PCAT and Tflash, or contact Thundercomm's technical support staff.

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